

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

We kick off the science section this month with the shifting magnetic north. Following that is an exploration of the Chang'e rover, after which we dive into what it means to be a planet. Right after that, we tell you why you can't have an Ant-Man suit, ever.



17 expressions for happiness

Scientists have studied how facial expressions are interpreted across cultures, and found that some emotions are expressed in more ways than others. While the human face can be contorted in 16,384 different ways, the researchers found only 35 different facial emotions across cultures. The researchers found that a large number of human expressions convey joy. <https://digit.in/17joy>

WHAT'S NEW

Electricity producing bacteria can be used for power generation

Some bacteria living in anoxic conditions, or an environment deprived of oxygen, can produce electricity as a waste product. These bacteria live in deep mines, or at the bottom lakes. The challenge has been to identify and isolate these bacteria. The bacteria are much smaller than mammalian cells, and are difficult to grow in a lab environment. Researchers from MIT have identified a new technique of isolating these useful bacteria, by segregating them on a property called polarizability, which has a high correlation with their ability to produce electricity. The traditional



approaches for identifying electricity producing bacteria was slow and time consuming. The researchers

from MIT were looking for a faster, less destructive way of isolating the useful bacteria. Developing the process for segregating the bacteria was painstaking, and took over ten years of research.

<https://digit.in/elec bact>

Saturn did not always have rings

Measurements from Cassini's final dive into Saturn have allowed researchers to precisely weigh the amount of material in the rings of Saturn. The material in the rings is roughly 40 percent of the mass of Mimas, a moon of Saturn. The measurement informs scientists that the rings were formed less than 100 million years ago, and maybe even as little as 10 million years ago. The finding shows that the ring



was certainly not formed 4.5 billion years ago. Essentially, the lower the mass, the younger the rings are. This is because when they were formed, they are made up of light and bright particles of ice. As the material

in the rings age, they become darker and heavier, as they accumulate contaminants and interplanetary debris. There are some signs that the material in the ring was dumped into orbit after a moon impacted with a comet. Mimas, which is in orbit in a gap in the rings, called the Cassini Division, has a large impact crater, known as the Herschel Crater which could have been the collision site.

<https://digit.in/sat rings>



Fossil fish shakes up family tree

The discovery of a 100 million year old fossilised fish has allowed researchers to get a better understanding of nearly all vertebrates, after finding this remote ancestor. The finding has led to the re-organisation of all fish and their descendants. <https://digit.in/hagfish>



Faster 3D printing with light

A new approach solidifies 3D shapes in their entirety, by shining light into a vat full of liquid. A pair of lights are used to create the shapes within liquid resin. The researchers have demonstrated the technique by printing a lattice, a boat, and the letter M. <https://digit.in/lightprint>



Probing clownfish anemone symbiosis

An unknown mechanism prevents the anemone from killing of the clownfish, while it consumes other fish in the sea. New research explores the effects of the exchange of bacteria between the two organisms, on their symbiotic relationship. <https://digit.in/clwnanmn>